A Bayesian Population-Dynamics Model for Clonal T Cells across Tissue and Phenotype in Glioblastoma

June 24, 2026

Contents

1 Setup

We observe annotated T cells from multiple patients, anatomical sites, and study timepoints. Each observed cell carries five labels: a patient, a TCR clonotype, a study timepoint, an anatomical site, and a discrete T-cell phenotype state.

The data are represented as aggregate clone-by-timepoint-by-site-by-phenotype counts, from which we construct clone-level observations across forward study steps.

1.1 Indices

We use the following indices throughout.

- Patients: $i = 1, \dots, I$.
- TCR clonotypes within patient i: $c = 1, \dots, C_i$.
- Shared study timepoints: $r \in \{1, \dots, R\}$, ordered as $\tau_1 < \tau_2 < \cdots < \tau_R$. The timepoint labels are comparable across patients.
- Anatomical site: $s \in \mathcal{S}$, with $\mathcal{S} = \{\mathrm{PBMC}, \mathrm{CSF}, \mathrm{Tumor}\}$.
- Annotated T-cell phenotype state: $k \in \{1, \dots, K\}$. In the current application $K = 11$, and all phenotypes share a single phenotype space.

1.2 Observed count tensor

For each patient i, clonotype c, timepoint r, site s, and phenotype k, let

$$Y_{icrsk} \in \mathbb{Z}_{\geq 0}$$

be the observed number of cells assigned to that combination: how many cells from patient i, clonotype c, and timepoint r were sampled from site s and assigned phenotype k. Collecting the site and phenotype axes,

$$Y_{icr} \in \mathbb{Z}_{\geq 0}^{|\mathcal{S}| \times K}$$

is the tissue-by-phenotype count tensor for clonotype c in patient i at timepoint r.

Each tissue sample also carries a sequencing depth and a missingness flag. For patient i, timepoint r, and tissue s, let

$$d_{i,r,s} \in \mathbb{Z}_{\geq 0}$$

be the per-sample sequencing depth: the VDJ cell count for the Cell Ranger run of that (patient, timepoint, tissue) sample. It is the Poisson exposure used below. Let

$$m_{i,r,s} \in \{0, 1\}$$

be the missingness indicator, equal to 1 when that sample was profiled and 0 otherwise.

1.3 Joint tissue–phenotype state space

The model operates on the joint tissue–phenotype state space

$$\mathcal{Z} = \mathcal{S} \times \{1, \ldots, K\}, \qquad z = (s, k),$$

where s is the tissue and k is the phenotype. Since $|\mathcal{S}| = 3$ and $K = 11$, the number of joint states is

$$L \;=\; |\mathcal{Z}| \;=\; |\mathcal{S}|\,K \;=\; 33.$$

The model therefore tracks clone counts across a 33-state tissue–phenotype space. When describing transitions we write the source state as $z = (a, u)$ and the destination state as $z' = (b, v)$, where $a, b \in \mathcal{S}$ are source and destination tissues and $u, v \in \{1, \ldots, K\}$ are source and destination phenotypes.

1.4 Valid forward steps

Forward steps are defined per patient. For patient i, let $\mathcal{R}_i \subseteq \{1, \ldots, R-1\}$ be the set of source timepoints at which patient i has a valid forward step. We index a step by its source timepoint $r \in \mathcal{R}_i$; the step runs from timepoint r to the next sampled timepoint, written $r+1$. Thus r is not a matrix index—it simply labels a forward step within patient i by where it starts.

Remark (Skipped timepoints)**.** Writing the destination as $r+1$ assumes consecutive sampling. When a patient is missing a timepoint—so that, say, T_1 and T_3 are paired—$r+1$ should be read as the *next available* timepoint for that patient. Which timepoints are paired is a data-preprocessing detail and does not change anything below.

Each $r \in \mathcal{R}_i$ is one observed forward step. Steps such as $T_1 \longrightarrow T_2$, $T_2 \longrightarrow T_3$, and $T_3 \longrightarrow T_4$ are treated as repeated observations of the *same* forward process: the growth matrix is shared across all patients and steps, and the indices i and r merely organize observations rather than selecting a separate matrix.

1.5 Clone-level observations within a step

Fix a patient i, a valid step $r \in \mathcal{R}_i$, and a clonotype c observed for that patient. The source and destination count vectors over the joint state space are read directly off the count tensor at timepoints r and $r+1$,

$$x_{irc}(s,k) = Y_{i\,c\,r\,s\,k}, \qquad y_{irc}(s,k) = Y_{i\,c\,(r+1)\,s\,k}, \qquad (s,k) \in \mathcal{Z},$$

so $x_{irc} \in \mathbb{Z}^L_{\geq 0}$ and $y_{irc} \in \mathbb{Z}^L_{\geq 0}$.

The source enters the model only after dividing out the source sequencing depth. The *depth-rescaled source* is

$$\tilde{x}_{irc}(z) = \frac{x_{irc}(z)}{d^{\mathrm{src}}_{ir,s}}, \qquad s = \mathrm{tissue}(z), \qquad d^{\mathrm{src}}_{ir,s} = d_{i,r,s},$$

i.e. each source count is divided by the source-timepoint depth of its tissue. It is treated as a fixed, known plug-in, not a random quantity: source abundance enters after dividing out source depth, and residual source-sampling variation is not modeled.

We retain clonotypes present at the *source* end of the step,

$$\mathcal{C}_{ir} = \left\{ c \in \{1, \ldots, C_i\} \; : \; N^{\mathrm{src}}_{irc} > 0 \right\}, \qquad N^{\mathrm{src}}_{irc} = \sum_{z \in \mathcal{Z}} x_{irc}(z).$$

A destination total of zero is now a modeled observation—a contraction or death signal—not an exclusion criterion. This gives the data hierarchy

$$i \;\longrightarrow\; r \in \mathcal{R}_i \;\longrightarrow\; c \in \mathcal{C}_{ir},$$

patients contain valid forward steps, and each forward step has a retained set of source-present clonotypes. A *clone-level forward observation* is the triple (i, r, c); for compact notation we write

$$j \;\equiv\; (i, r, c), \qquad x_j = x_{irc}, \qquad \tilde{x}_j = \tilde{x}_{irc}, \qquad y_j = y_{irc}.$$

The destination vector y_j keeps its per-tissue structure: each tissue sub-block $\{(s', k) : k = 1, \ldots, K\}$ of y_j carries its own destination depth $d_{j,s'} = d_{i,r+1,s'}$ and missingness flag $m_{j,s'} = m_{i,r+1,s'}$.

1.6 Two kinds of destination zero

Destination zeros are part of the likelihood specification, and the two kinds are treated differently.

1. **Missing sample** ($m_{j,s'} = 0$). The tissue sub-block $\{(s', k)\}$ was not profiled at the destination timepoint. It is excluded from the likelihood entirely—masked out, never entered as a zero count.
2. **Depth-zero** ($m_{j,s'} = 1$, clone not captured). A genuine observed zero, modeled natively by the Poisson likelihood below: a small mean $d_{j,s'}\, \mu_j(z')$ emits zero with positive probability while $\mu_j(z') > 0$. No zero-inflation component is added.

The depth-zero mechanism is what permits a clone to be absent at one step and present at the next.

1.7 Setup summary

The setup produces the clone-level forward observations

$$\mathcal{D} = \left\{ (\tilde{x}_j,\, y_j,\, d_j,\, m_j,\, i_j,\, r_j,\, c_j) \right\}_{j=1}^{J}, \qquad j \equiv (i_j, r_j, c_j),$$

each carrying a depth-rescaled source vector, a destination count vector, the per-tissue destination depths $d_j = (d_{j,s'})_{s' \in \mathcal{S}}$ and missingness flags $m_j = (m_{j,s'})_{s' \in \mathcal{S}}$, and a patient / step / clonotype identity. The count vectors live in the joint tissue–phenotype state space $\mathcal{Z} = \mathcal{S} \times \{1, \dots, K\}$, with $|\mathcal{Z}| = 33$.

1.8 Notation summary

Table 1 collects the core modeling symbols; the variational quantities used only for fitting (the Gamma parameters $\mathrm{shape}_{zz'}, \mathrm{rate}_{zz'}$, the responsibilities $r_{jzz'}$, and ψ) are introduced where they are used in the inference section.

Symbol	Meaning
$i = 1, \dots, I$	patient index
$c = 1, \dots, C_i$	TCR clonotype within patient i
$r \in \{1, \dots, R\}$	shared study timepoint, $\tau_1 < \cdots < \tau_R$
$s \in \mathcal{S}$	anatomical site, $\mathcal{S} = \{\mathrm{PBMC}, \mathrm{CSF}, \mathrm{Tumor}\}$
$k \in \{1, \dots, K\}$	phenotype state ($K = 11$)
$z = (s, k) \in \mathcal{Z}$	joint tissue–phenotype state ($L = \lvert\mathcal{Z}\rvert = 33$)
Y_{icrsk}	observed count tensor
$d_{i,r,s}$	per-sample sequencing depth (Poisson exposure)
$m_{i,r,s} \in \{0, 1\}$	sample missingness indicator
$\mathcal{R}_i$	valid forward steps for patient i ($\subseteq \{1, \dots, R-1\}$)
$r \in \mathcal{R}_i$	forward step, from source timepoint r to $r+1$
C_{ir}	clonotypes present at the source end of step r
$j = (i_j, r_j, c_j)$	clone-level observation, $j = 1, \dots, J$
$x_j,\ y_j \in \mathbb{Z}^L_{\geq 0}$	source / destination count vectors
$\tilde{x}_j$	depth-rescaled source, $\tilde{x}_j(z) = x_j(z)/d^{\mathrm{src}}_{j,s}$
$M \in \mathbb{R}^{L \times L}_{\geq 0}$	non-negative growth matrix (free row sums)
$\mu_j = \tilde{x}_j M \in \mathbb{R}^L_{\geq 0}$	destination intensity vector
$a_z,\ b_z$	Gamma prior shape / rate for row z
$w_{jzz'}$	backward attribution probability
$\xi_{jzz'}$	latent allocation count (computational)
$\mathcal{D}$	full dataset

Table 1: Core modeling notation.

2 Joint Tissue–Phenotype Growth Matrix

The central latent object is a single joint tissue–phenotype *growth matrix*

$$M \in \mathbb{R}^{L \times L}_{\geq 0}, \qquad L = 33,$$

whose rows are source states and whose columns are destination states. For source state $z = (a, u)$ and destination state $z' = (b, v)$, the entry $M_{zz'}$ is the expected number of destination-z' cells produced per source-z cell over one forward step (one 8-week interval):

$$M_{zz'} = M_{(a,u),(b,v)} = \mathbb{E}[\text{ cells in state } (b, v) \text{ at } r + 1 \text{ per cell in state } (a, u) \text{ at } r\ ].$$

2.1 Non-negative rows and growth factors

M has non-negative entries, $M_{zz'} \geq 0$, and *free* row sums—rows are not constrained to sum to one. The row sum

$$g_z = \sum_{z' \in \mathcal{Z}} M_{zz'}$$

is the net 8-week growth factor of source state z: $g_z > 1$ is net expansion of that state, $g_z < 1$ net contraction or death.

2.2 Destination intensity vector

For observation j with depth-rescaled source $\tilde{x}_j$, the destination *intensity vector* is the push-forward of $\tilde{x}_j$ through M:

Central growth calculation

$$\mu_j = \tilde{x}_j M \in \mathbb{R}^L_{\geq 0}, \qquad \text{equivalently} \qquad \mu_j(z') = \sum_{z \in \mathcal{Z}} \tilde{x}_j(z)\, M_{zz'}.$$

The expected destination count in state z' scales the intensity by the destination depth of its tissue,

$$\mathbb{E}\big[y_j(z') \mid M\big] = d_{j,s'}\, \mu_j(z'), \qquad s' = \text{tissue}(z').$$

There is no simplex constraint on μ_j and no conditioning on a destination total.

2.3 Role of forward steps

The same matrix M is used for every patient i and step $r \in \mathcal{R}_i$. The pair (i, r) selects which source and destination vectors to build, not a different matrix; all steps are repeated observations of a common one-step forward process.

3 Bayesian Reading of the Growth Matrix

The matrix M is a latent non-negative kernel of per-cell destination intensities over the joint tissue–phenotype state space.

3.1 Forward generative reading

Read forward, the model is a Poisson superposition. Treating the depth-rescaled source as a population with $\tilde{x}_j(z)$ cells in each state z, each source cell in state z independently emits a Poisson-distributed number of cells into each destination state z' with mean $M_{zz'}$. Summing these emissions over all source states and scaling by the destination depth, the destination-z' count is, by Poisson superposition,

$$y_j(z') \mid M \;\sim\; \mathrm{Poisson}\Big(d_{j,s'} \sum_{z\in\mathcal{Z}} \tilde{x}_j(z)\, M_{zz'}\Big) = \mathrm{Poisson}\big(d_{j,s'}\, \mu_j(z')\big).$$

3.2 Likelihood for destination counts

Observation likelihood

$$y_j(z') \;\sim\; \mathrm{Poisson}\big(d_{j,s'}\, (\tilde{x}_j M)(z')\big), \qquad s' = \mathrm{tissue}(z'),$$

evaluated only over non-missing destination sub-blocks ($m_{j,s'} = 1$). The latent matrix enters the likelihood only through the intensity $\mu_j = \tilde{x}_j M$.

3.3 Prior over growth-matrix entries

Each entry receives an independent Gamma prior in shape–rate form,

$$M_{zz'} \;\sim\; \mathrm{Gamma}(a_z, b_z),$$

with row-specific shape a_z and rate b_z.

Remark (Support restrictions are an implementation detail)**.** In practice the implementation may exclude biologically implausible transitions by fixing the corresponding entries of M to zero (placing the Gamma prior only on the retained entries of each row). This sparsifies M but does not change the form of the model; sums over destination states below are read over the retained entries. We treat the support as a fixed preprocessing choice.

For the likelihood to support the data, the support guard is: for every non-missing destination state z' with $y_j(z') > 0$, at least one source state in the support of $\tilde{x}_j$ must reach z' (i.e. have $M_{zz'} > 0$).

3.4 Posterior over growth matrices

After observing the data $\mathcal{D}$, inference targets the posterior $p(M \mid \mathcal{D})$, which captures uncertainty about the joint tissue–phenotype growth kernel after all clone-level forward observations. Posterior summaries include posterior means $\mathbb{E}[M_{zz'} \mid \mathcal{D}]$ and credible intervals for individual entries.

3.5 Posterior predictive destination counts

Because M is latent, the intensity $\mu_j = \tilde{x}_j M$ is also uncertain. Posterior draws of M induce replicated destination counts

$$y_j^{\mathrm{rep}}(z') \;\sim\; \mathrm{Poisson}\big(d_{j,s'}\, \mu_j(z')\big)$$

over the non-missing sub-blocks ($m_{j,s'} = 1$).

3.6 Backward attribution

The reverse question—which source state produced an observed destination cell—is a posterior attribution. For observation j,

$$w_{jzz'} = \frac{\tilde{x}_j(z)\, M_{zz'}}{\sum_{\tilde{z}\in\mathcal{Z}} \tilde{x}_j(\tilde{z})\, M_{\tilde{z}z'}} = \frac{\tilde{x}_j(z)\, M_{zz'}}{\mu_j(z')},$$

the same form as the forward intensity with source and destination roles reversed. Because the destination counts are Poisson, these probabilities also induce a multinomial decomposition of each observed count $y_j(z')$ into latent source-to-destination allocation counts. Those counts are used only for fitting, so we introduce them together with the inference algorithm in Section 5.

4 Full Generative Story

The full model is a model for the one-step expansion and redistribution of clone mass across tissue–phenotype states. It can be summarized as follows.

Generative model

1. **State space.** Define $\mathcal{Z} = \mathcal{S} \times \{1, \ldots, K\}$ with $|\mathcal{Z}| = 33$. Each state $z = (s, k)$ pairs an anatomical tissue with an annotated T-cell phenotype.
2. **Growth-matrix entries.** For each source state z and destination state z', draw independently
$$M_{zz'} \sim \mathrm{Gamma}\big(a_z, b_z\big).$$
Entry $M_{zz'}$ is the expected number of destination-z' cells per source-z cell over one forward step.
3. **Source.** For each patient i, step $r \in \mathcal{R}_i$, and retained clonotype $c \in \mathcal{C}_{ir}$, form the observation $j = (i, r, c)$ and the fixed depth-rescaled source $\tilde{x}_j$.
4. **Intensity.** Push the source forward, $\mu_j = \tilde{x}_j M$.
5. **Observation.** For each non-missing destination sub-block ($m_{j,s'} = 1$), draw the destination counts
$$y_j(z') \sim \mathrm{Poisson}\big(d_{j,s'}\, \mu_j(z')\big), \qquad s' = \mathrm{tissue}(z').$$

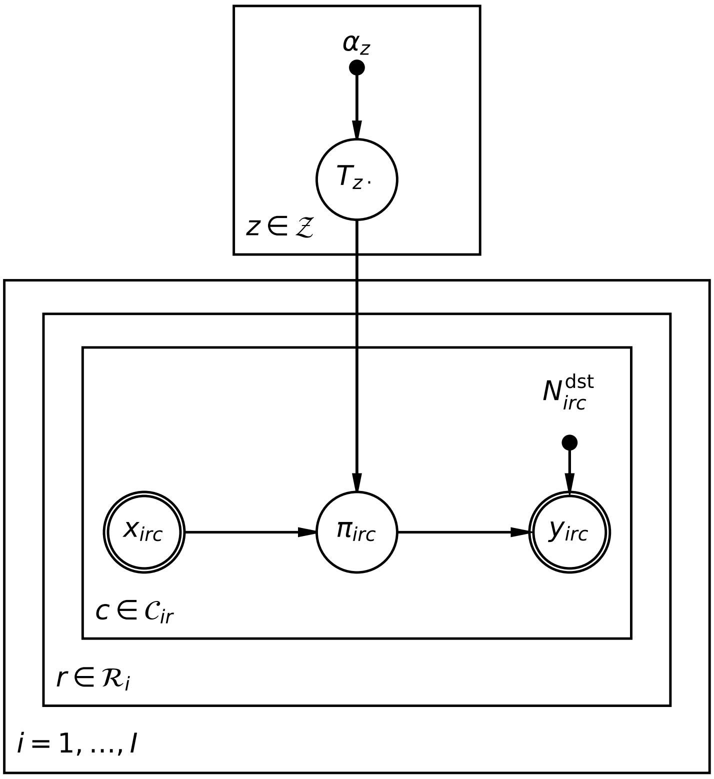
Putting the pieces together, the generative model is, compactly,

$$M_{zz'} \sim \mathrm{Gamma}\big(a_z, b_z\big), \qquad \mu_j = \tilde{x}_j M, \qquad y_j(z') \sim \mathrm{Poisson}\big(d_{j,s'}\, \mu_j(z')\big) \quad (m_{j,s'} = 1),$$

with the corresponding dependency structure shown in Figure 1.

5 Inference

Inference estimates the posterior over the shared growth matrix M. The observed data are the clone-level forward observations $\mathcal{D}$; the depth-rescaled sources $\tilde{x}_j$, the depths $d_{j,s'}$, and the missingness flags $m_{j,s'}$ are fixed inputs, as are the Gamma prior parameters a_z, b_z.



Node	**Meaning**
x_{irc}	observed source counts at the start of the step (over the 33 states)
$\theta_{irc} = \mathrm{Normalize}(x_{irc})$	source distribution (deterministic; folded into the $x \to \pi$ edge)
$\pi_{irc} = \theta_{irc} T$	model-implied destination distribution
y_{irc}	observed destination counts at the end of the step
$T_{z\cdot}$	latent transition row, shared across all patients and steps
N_{irc}^{dst}	destination cell total (fixed)
α_z	Dirichlet prior concentration (fixed)

Figure 1: Plate diagram of the generative model (node key above). The transition rows $T_{z\cdot}$ are drawn once per source state z (top plate) from a Dirichlet prior α_z; this block sits *outside* the patient plate, so a single T is shared across all patients and steps. The observation side nests patient i, forward step $r \in \mathcal{R}_i$, and retained clonotype $c \in \mathcal{C}_{ir}$, so a clone-level observation is the triple (i, r, c). Within the innermost plate the observed source counts x_{irc} are normalized and pushed through T to the destination distribution $\pi_{irc} = \theta_{irc} T$, which then drives a multinomial draw of the observed destination counts y_{irc} with total N_{irc}^{dst}. Double circles (x_{irc}, y_{irc}) mark observed quantities and solid dots (α_z, N_{irc}^{dst}) mark fixed inputs; the deterministic normalization $\theta_{irc} = \mathrm{Normalize}(x_{irc})$ is folded into the $x \to \pi$ edge. Latent allocation counts used for fitting are a computational augmentation introduced in Section 5.

5.1 Learned quantity

The main learned object is the posterior $p(M \mid \mathcal{D})$. A single-entry summary is

$$\mathbb{E}\big[M_{(a,u),(b,v)} \mid \mathcal{D}\big]\,,$$

the posterior mean number of destination-(b,v) cells produced per source-(a,u) cell over one forward step.

5.2 Variational approximation

We approximate $p(M \mid \mathcal{D})$ by mean-field variational inference. The difficulty is that an observed destination count in state z' is a *superposition* of contributions from every source state z, and these source-to-destination allocations are unobserved. Variational inference augments the model with latent allocation counts and fits a factorized approximation by coordinate ascent.

5.3 Latent allocation counts

Let $\xi_{jzz'}$ be the unobserved number of destination-z' cells in observation j contributed by source state z. By Poisson superposition each source contribution is itself Poisson, $\xi_{jzz'} \sim \mathrm{Poisson}\big(d_{j,s'}\,\tilde{x}_j(z)\,M_{zz'}\big)$ independently, and they decompose each observed destination count,

$$\sum_{z\in\mathcal{Z}} \xi_{jzz'} = y_j(z').$$

Conditional on the total $y_j(z')$, the allocations are multinomial with probabilities equal to the backward attribution $w_{jzz'}$ of Section 3.6; variational inference replaces these with an approximate responsibility $r_{jzz'}$.

5.4 Variational family

We adopt a mean-field factorization $\mathfrak{q}(M,\xi) = \mathfrak{q}(M)\,\mathfrak{q}(\xi)$. The growth matrix receives independent Gamma factors, one per entry,

$$\mathfrak{q}(M) = \prod_{z\in\mathcal{Z}}\prod_{z'\in\mathcal{Z}} \mathrm{Gamma}\big(M_{zz'};\ \mathrm{shape}_{zz'},\ \mathrm{rate}_{zz'}\big),$$

and the allocation variables receive multinomial factors (a Poisson vector conditioned on its total),

$$\mathfrak{q}(\xi) = \prod_{j=1}^{J}\ \prod_{z'\,:\,m_{j,s'}=1} \mathrm{Multinomial}\big(\xi_{j\cdot z'};\ y_j(z'),\ r_{j\cdot z'}\big),$$

where $r_{jzz'}$ is the variational responsibility assigning destination state z' in observation j to source state z.

5.5 Coordinate updates

The algorithm alternates between updating the allocation responsibilities $r_{jzz'}$ and the Gamma parameters $(\mathrm{shape}_{zz'}, \mathrm{rate}_{zz'})$.

5.5.1 Allocation update

For each observation j, destination state z' with $m_{j,s'} = 1$, and source state z,

$$r_{jzz'} \propto \tilde{x}_j(z)\,\exp\!\big(\mathbb{E}_{\mathfrak{q}}[\log M_{zz'}]\big),$$

normalized over source states so that $\sum_{z\in\mathcal{Z}} r_{jzz'} = 1$ for each j and z' (the common depth $d_{j,s'}$ cancels in the normalization). Under the Gamma factor, the required expectation is

$$\mathbb{E}_{\mathfrak{q}}[\log M_{zz'}] = \psi(\mathrm{shape}_{zz'}) - \log(\mathrm{rate}_{zz'}),$$

where ψ is the digamma function.

5.5.2 Entry update

For each source state z and destination state z', the Gamma–Poisson conjugate update is

$$\mathrm{shape}_{zz'} = a_z + \sum_j y_j(z')\,r_{jzz'}, \qquad \mathrm{rate}_{zz'} = b_z + \sum_j d_{j,s'}\,\tilde{x}_j(z),$$

where both sums run over observations whose destination sub-block s' is non-missing ($m_{j,s'} = 1$). The shape accumulates the expected number of $z \to z'$ cells allocated from the data; the rate accumulates the depth-weighted source exposure.

5.6 Optimization objective

The coordinate updates maximize the evidence lower bound (ELBO)

$$\mathcal{L}(\mathfrak{q}) = \mathbb{E}_{\mathfrak{q}}[\log p(M,\xi,\mathcal{D})] - \mathbb{E}_{\mathfrak{q}}[\log \mathfrak{q}(M,\xi)]\,,$$

and the algorithm iterates between the allocation update and the entry update until $\mathcal{L}(\mathfrak{q})$ stabilizes.

5.7 Estimated growth matrix

After convergence, the fitted growth matrix is the variational posterior mean of each Gamma factor,

$$\widehat{M}_{zz'} = \mathbb{E}_{\mathfrak{q}}[M_{zz'}] = \frac{\mathrm{shape}_{zz'}}{\mathrm{rate}_{zz'}},$$

explicitly *not* normalized across destinations z'. The matrix $\widehat{M}$ is the estimated one-step growth matrix over the 33-state tissue–phenotype space.

5.8 Algorithm summary

Variational algorithm (CAVI)

1. **Initialize** the variational Gamma parameters $(\mathrm{shape}, \mathrm{rate})$.
2. **Repeat until the ELBO $\mathcal{L}(\mathfrak{q})$ converges:**

- update the allocation responsibilities $r_{jzz'} \propto \tilde{x}_j(z)\,\exp\!\big(\mathbb{E}_q[\log M_{zz'}]\big)$;
- update the entry parameters $\text{shape}_{zz'} = a_z + \sum_j y_j(z')\, r_{jzz'}$, $\text{rate}_{zz'} = b_z + \sum_j d_{j,s'}\, \tilde{x}_j(z)$.

3. **Return** the posterior mean $\widehat{M} = \mathbb{E}_q[M]$.

6 Posterior Summaries

With the variational posterior fitted and $\widehat{M}$ in hand, the following quantities are read off it.

6.1 Trafficking and growth

The fitted matrix $\widehat{M}$ admits two readings, on its off-diagonal tissue blocks and on its row sums.

Trafficking. For source state $z = (a, u)$ and a destination tissue $b \neq a$, the off-diagonal block sum

$$\tau_{z \to b} = \sum_{v=1}^{K} \widehat{M}_{(a,u),(b,v)}$$

is the expected number of progeny of a source-z cell that land in tissue b over the step; summing over destination tissues $b \neq a$ gives the total out-of-tissue trafficking from state z.

Growth. The full row sum (diagonal block included)

$$g_z = \sum_{z' \in \mathcal{Z}} \widehat{M}_{zz'}$$

is the net per-cell progeny count of source state z over the step.

Off-diagonal block sums measure trafficking; the row sum, with the diagonal included, measures net growth.

6.2 Rate decomposition

To express the step as per-week rates, define the generator

$$B = \frac{1}{\Delta t} \log \widehat{M}, \qquad \Delta t = 8 \text{ weeks},$$

the matrix logarithm of $\widehat{M}$ divided by the step length. The off-diagonal entries $B_{zz'} \geq 0$ ($z' \neq z$) are the per-week trafficking rates between states, and the row sums of B are the net per-capita growth rates per week. As above, the off-diagonal entries carry trafficking and the row sums carry growth.